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VIA E-MAIL (ostp-ai-rfi@nitr.gov)

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RE: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dear Mr. D'Souza

Eli Lilly and Company (Lilly) appreciates the opportunity to provide comments on the above-referenced Request for Information (RFI).¹ Lilly is one of the country's leading innovation-driven, research-based pharmaceutical and biotechnology corporations. Our company is devoted to seeking answers for some of the world's most urgent medical needs through discovery and development of breakthrough medicines. Ultimately, our goal is to develop products that save and improve patients' lives. We recognize the tremendous potential of artificial intelligence (AI) to transform the healthcare landscape, and we are eager to contribute to shaping an AI policy framework that fosters innovation while safeguarding public health, security, and privacy. Our comments below reflect our focus on leveraging AI to drive biomedical advancement and improve public health outcomes.

National Security

Lilly recognizes the critical intersection of biopharmaceuticals, AI, and national security, particularly in addressing emerging AI-driven biosecurity threats. As AI continues to evolve, it is imperative that federal regulators take proactive steps to safeguard public health, while fostering innovation in the biopharmaceutical sector. Lilly believes that strengthening protections for the domestic biopharmaceutical industry enhances the nation's ability to counter emerging biosecurity risks, while reinforcing its leadership in global health security. Moreover, AI-driven synthetic biology threats pose unique challenges that require both public and private sectors to work together in developing robust strategies to mitigate risks and maintain national resilience. Below, we provide key recommendations to strengthen national security and mitigate risks associated with AI-driven biological threats:

- **Regulators should designate key biopharmaceutical manufacturers as national security assets to protect against foreign influence and ensure stability in public health preparedness** – Lilly supports formally classifying select pharmaceutical companies as national security assets, acknowledging their critical role in national defense and emergency response. This designation would provide protections against foreign takeovers or undue influence, particularly from nations whose interests may conflict with U.S. security priorities. Safeguarding

¹ National Science Foundation, Request for Information on the Development of an Artificial Intelligence (AI) Action Plan. 90 Fed. Reg. 9088, 9089 (Feb. 6, 2025).

domestic biopharmaceutical companies would strengthen the nation's ability to respond to health emergencies and mitigate AI-driven biosecurity risks.

- **Pharmaceutical manufacturing facilities should be classified as critical infrastructure to enhance security against cyberattacks, terrorism, and AI-driven biological threats** – Federal agencies should establish policies that categorize pharmaceutical manufacturing sites as critical infrastructure, ensuring they receive enhanced security measures and protections. This classification would enable increased investment in cybersecurity defenses, emergency preparedness programs, and risk mitigation strategies against AI-driven threats, such as synthetic biology misuse. By formally recognizing the pharmaceutical sector as part of critical infrastructure, regulators can help protect the industry from emerging risks and maintain national health security.
- **The AI Action Plan should include directives to establish a national biosecurity strategy to strengthen the nation's preparedness for AI-driven biological threats** – Lilly supports establishing a comprehensive national biosecurity strategy that outlines clear protocols for responding to AI-generated biological threats, including the potential use of AI in developing synthetic pathogens. This strategy should include collaboration between government agencies, the biopharmaceutical industry, and national defense organizations to ensure a coordinated response to emerging threats. Implementing a forward-looking regulatory framework would ensure federal agencies can bolster national resilience against AI-driven biosecurity risks while fostering innovation in the biopharmaceutical sector.

Data Privacy and Cybersecurity

As AI becomes increasingly embedded in health care operations, data privacy and cybersecurity frameworks should evolve to balance innovation with robust protections for personal health information. Lilly supports targeted, sector-specific updates to privacy regulations that reflect the opportunities and challenges posed by AI, ensuring continued medical innovation while safeguarding sensitive data. To this end, we recommend the following:

- **Regulators should update HIPAA regulations and other health privacy frameworks to address AI-driven technologies that process personal health information** – Federal agencies should modernize existing health privacy regulations to ensure that AI applications handling personal health data are appropriately regulated. This includes clarifying how AI technologies interact with HIPAA protections and establishing guidelines that allow for responsible AI-driven research in drug discovery and clinical trials while maintaining strong patient privacy standards. Lilly believes that refining privacy laws to account for AI's role in health care would support both innovation and consumer trust in AI-driven medical advancements.
- **Cybersecurity standards should be tailored to AI-driven biopharmaceutical research and manufacturing to mitigate the risks of cyber threats and data breaches** – Federal agencies should introduce cybersecurity guidelines specifically designed to protect AI-driven research, development, and manufacturing in the biopharmaceutical industry. These guidelines should promote the adoption of advanced cybersecurity technologies, such as encryption and AI-powered threat detection, while ensuring that regulations remain adaptable to emerging threats. Targeted cybersecurity requirements can help protect sensitive health data and prevent cyberattacks on pharmaceutical infrastructure.

Research and Development (R&D)

Lilly supports the continued growth in the use of AI to drive meaningful advances in drug discovery and development, clinical trials, and regulatory review, and we suggest that federal regulators' efforts focus on creating an environment that encourages innovation while safeguarding public health. We recommend that any regulatory actions related to AI technologies prioritize providing clarity, flexibility, and standardized approaches to facilitate the development and approval of life-saving treatments, especially for patients with rare or underserved conditions. Below we provide additional recommendations related to the use of AI in R&D and clinical trials:

- **Regulatory guidance should spur the use of innovative approaches to development while ensuring that new technologies are properly validated and monitored** – Drugs discovered and developed leveraging AI technology should continue to meet the same safety and efficacy standards as drugs developed with traditional methods. As also described in the recent FDA Guidance "[Considerations for the Use of Artificial Intelligence To Support Regulatory Decision-Making for Drug and Biological Products](#)", AI algorithms should be free from bias and avoid exacerbating health disparities. The integration of AI technology and good manufacturing processes into drug discovery, research, and development brings promising benefits to patients, yet its potential continues to emerge.

Therefore, any actions to regulate AI technology should consider the implications of jeopardizing innovation within the full drug lifecycle. Federal regulators should seek to provide regulatory certainty and adoption of standardized approaches for the use of AI technologies to maintain a gold standard of safety and efficacy as well as help to encourage the appropriate adoption of AI within drug discovery and R&D.

- **Regulators should support the potential for AI to meaningfully advance medical discoveries for rare disease and disease with unmet medical need** – AI has the potential to promote efficiencies and increase the availability of treatments, especially for patients battling rare diseases and those with unmet medical needs. AI's capabilities in early drug discovery, particularly in identifying new therapeutic targets, could revolutionize the way we approach medical challenges in underserved therapeutic areas.
- **Regulatory guidance and frameworks should provide greater clarity and predictability regarding acceptance of the use of AI within clinical trials** – Lilly believes that AI algorithms used in clinical trials must be validated during the FDA regulatory review process to mitigate the risk of bias in data and to ensure the integrity of clinical outcomes. As drug developers continue to enhance the representativeness of trial data sets, AI tools can be used to mitigate algorithmic bias and improve health outcomes if used appropriately.

However, a challenging aspect to leveraging AI in clinical trials is understanding whether the data sets used by or to inform AI algorithms are comprehensive and non-biased. Additionally, developers need certainty from regulators regarding their acceptance of AI tools within clinical study, which ideally will include a recognition that regulators will remain flexible as the use of AI in clinical trials continues to evolve.

Intellectual Property and Innovation

Lilly supports the use of AI technology in developing innovative, patentable medicines, which accelerates medical innovation, and believes that intellectual property (IP) protections should continue to reward human creativity. Additionally, Lilly advocates for clearer criteria around IP

protections for medicines developed using AI, as strong and clear IP safeguards are critical to sustaining the pace of biomedical discovery and ensuring the continued advancement of life-saving treatments. To this end, we recommend the following:

- **The USPTO guidance position that AI-assisted inventions are not categorically unpatentable for improper inventorship should be maintained** – IP protections for AI-assisted innovation should continue to appropriately reward human creativity that remains fundamental to AI-assisted medical innovation, thus promoting the progress of science and improving the lives of patients.
- **Regulators should clarify the criteria for IP protections for medicines using AI technology during discovery** – IP protections for AI technology need clarity and should remain strong as IP is the backbone of continued drug discovery and advancements into biomedical innovation. Without strong and sufficient protections for IP, the biomedical industry could not continue at its historic pace of bringing treatments and cures to patients as quickly as possible.

Lilly strongly supports the advancement of AI technologies within the biopharmaceutical industry, particularly in drug discovery, research, and development. We recognize that the intersection of AI with public health, national security, and IP requires thoughtful, coordinated action from federal agencies, private industry, and academic institutions. As AI continues to evolve, it is imperative that policymakers create regulatory frameworks that encourage innovation while safeguarding privacy, national security, and fairness. Lilly looks forward to continued collaboration with the NITRD and other stakeholders to build a future where AI can be fully leveraged to benefit patients, strengthen public health, and drive biomedical breakthroughs. Thank you for your consideration of these important issues.

Sincerely,

Diogo Rau
Executive Vice President and Chief Information and Digital Officer
Eli Lilly and Company

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